

SleepDock - Smart Alarm Clock

Josh Kieler
CADD240 - DE10

Meet the SleepDock



SleepDock is a compact bedside device that combines a smart speaker and magnetic wireless phone dock into one simple solution. Users just place their phone on the dock and wake up to a cleaner, more organized nightstand experience.

SleepDock - Prototype BOM

<u>Item #</u>	<u>SubSystem</u>	<u>Description</u>	<u>QTY</u>	<u>Price</u>
1	COMPUTE	RaspberryPI 5	1	\$55.00
2	COMPUTE	MicroSD Card	1	\$15.00
3	USER I/O	capacitive touch display	1	\$58.00
4	VOICE	4-mic far-field array	1	\$30.00
5	AUDIO	40mm speaker 5 watt	1	\$5.00
6	AUDIO	D Amplifier Breakout - MAX98357A	1	\$6.00
7	CHARGER	MAG SAFE PUCK	1	\$12.00
Estimated Cost				\$181.00

The Competition

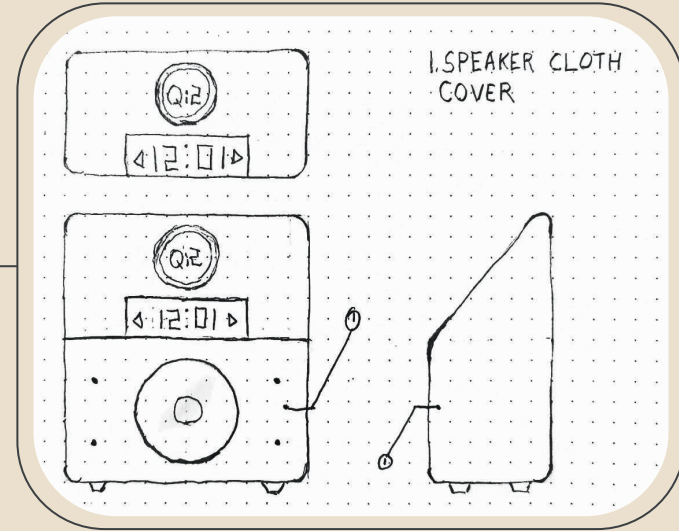
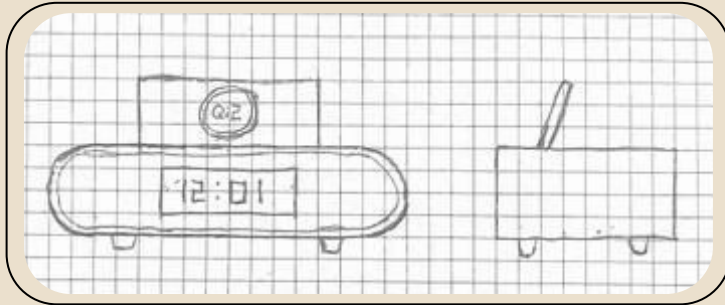
- Under powered charger
- No integrated smart features
- Overly complex designs, not fumbling in the dark friendly
- No voice control



The Idea

Modern Smart Speaker Form

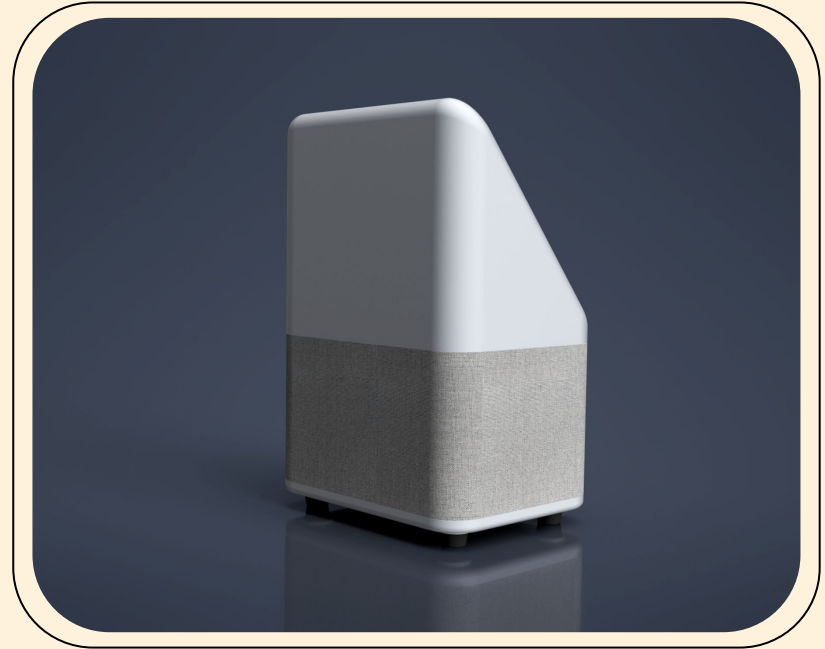
- Simple
- Congruent aesthetic to modern smart appliances
- Preferred by 75% of Design Team



Retro Clock Radio Form

- Inspired by classic clock radio
- Classic footprint
- Charger looks tacked on
- Oldest member of the team preferred this design

Front/Back View



Display Layout

Phone Charge Pad

Qi2/MagSafe compatible magnetic charging pad up to 18W.

Capacitive Display

Displays the Date/Time alarm condition in stand alone mode. Eliminates physical buttons.

Context Specific Controls

Controls adapt to user need. Whole display can be snooze button.

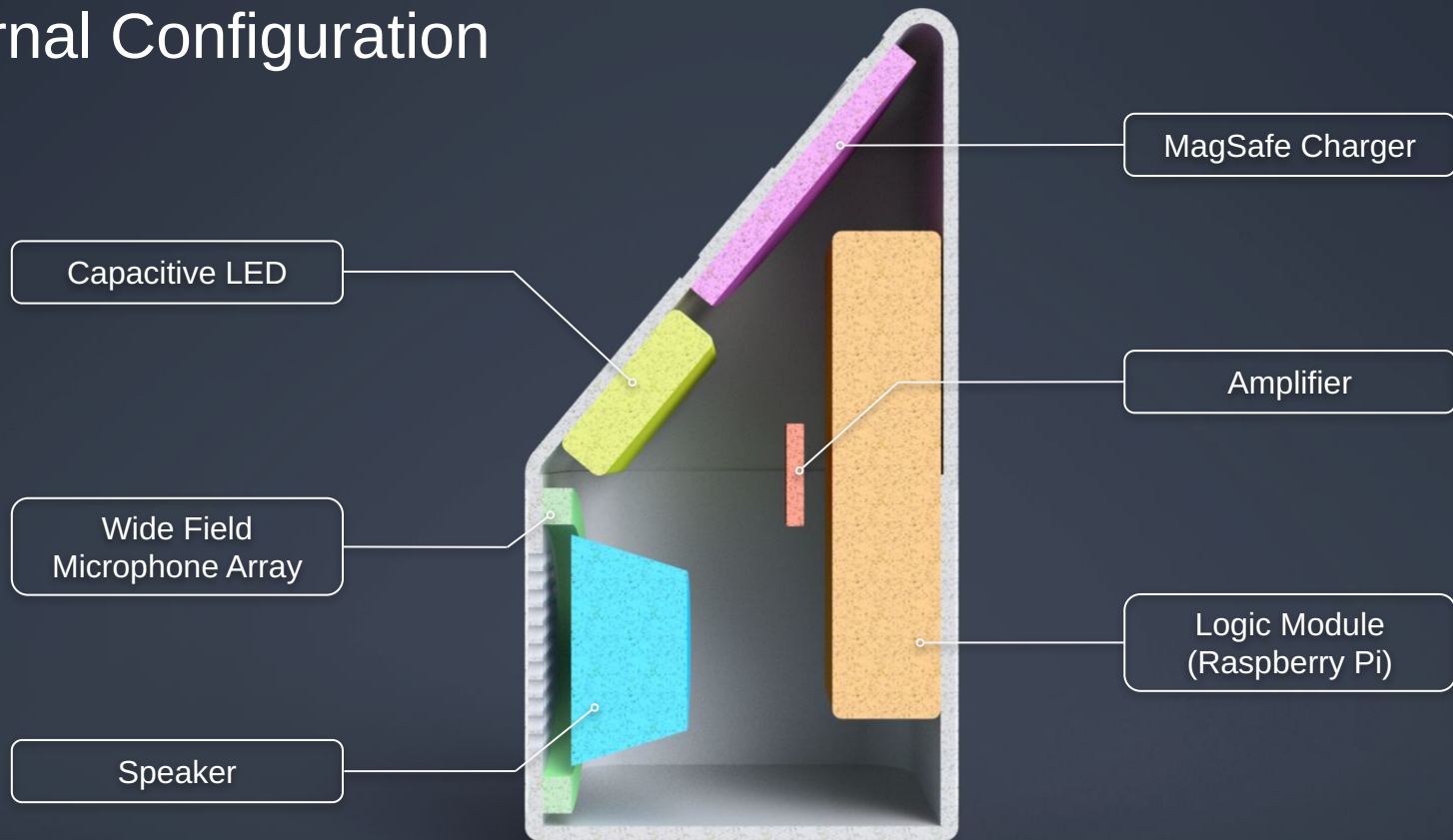




Key Features

- 18W charging via magsafe pad
- stand alone clock display
- 18W - 2 inch speaker
- Wide field microphone array
- Smart Assistant Capable Logic Board
- Smartphone display/operation via app
- Capacitive controls for sleek form factor

Internal Configuration





Questions?

Josh Kieler